



CS 03500 - Foundations of Cybersecurity

Project Part 4: System Hardening and Auditing

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Date: 10/21/2024

Firstly, I logged on to windows 10 system using administrative account, after successfully logging on, I opened the antivirus system installed on the system, McAfee, navigated to account tab and selected update app. This tab provides the option to check for updates now and states when the system was last updated. According to the McAfee software installed, the last update was Monday, October 21, 2024, at 6:10 pm. This means that the antivirus software running is up to date as of earlier today. In addition, McAfee antivirus software allows users to set up their own scheduled scans, as this system is slated to automatically scan the system every Sunday at 8:00 am. Also, the last scan was completed on Sunday October 20, 2024, and no significant threat was detected after the completion of the last scan. The next scan will be completed on Sunday 27, 2024 at 8:00 am. This scheduling timeslot was primarily selected to ensure that the scan does not slow systems down during peak business hours.

Secondly, I launched Local Group Policy Editor to begin configuring audit logging to record all failed password attempts into the system. I then navigated through the following path in the left-hand pane: Local Computer Policy > Computer Configuration > Windows Settings > Security Settings > Local Policies > Audit Policy. In the right-hand pane, I located the “Audit logon events” settings, then double click on it to access the configuration settings and selected “Failure” under audit these attempts, and then clicked OK to ensure changes were saved. These configurations will record failed logon events which can be viewed on Event Viewer. In the Event Viewer window, navigated to Windows Log > Security to identify failed login attempts which are listed as Event ID 4625.

Thirdly, after checking the installed antivirus software and configuring audit logging, I then looked for potentially problematic security events in the last 24 hours. Upon examining the results from the antivirus scan, it was concluded that there was no significant threat. However, the audit log recorded three failed logon attempts in the past 24-hour period. After a thorough investigation into these unsuccessful logon attempts, it was discovered that there were errors from employees who mistyped their password. While no malicious intent was suspected from these logon attempts, it remains a concern of security for the company because attackers can attempt to logon with user credentials to gain unauthorized access. Therefore, continuous monitoring of the audit logging files remains an integral part in protecting the company against attackers.

To correct the problem encountered during the 24-hour monitoring, I quickly identified that there were three failed login attempts. Upon investigation, it was discovered that there was a mistyped password from an employee. In this case, I recommend that Multi-Factor Authentication (MFA) be implemented for all employees to secure all users' accounts and prevent unauthorized access to the company network. In addition to MFA, I requested that employees use complex passwords and always safeguard their password. These strategies will reduce the risk of unauthorized access. Moreover, I would recommend that antivirus scans be scheduled daily as a safeguarding tool to detect and quarantine any threat found. This schedule should take place in off peak hours when less users are active.

System hardening is an essential cybersecurity strategy developed to enhance the company defense against cyber threats. It involves using various techniques, tools, and methods to reduce vulnerabilities in servers and computers by disabling unnecessary services, removing unused applications, and enabling built-in security features. By minimizing potential attack surfaces, system hardening limits opportunities for cyber attackers to compromise your IT environment. In addition to protecting your data and systems from cyber threats, system hardening is essential for meeting legal and regulatory compliance. By adopting strong security measures and hardening techniques, organizations can align with industry standards and regulatory guidelines. This approach not only helps avoid legal penalties and fines but also fosters trust with customers and partners, showcasing the organization's dedication to data protection and cybersecurity.

Work Cited

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